

PICU Data Collaborative

WFPICCS Schema v1.1

Summary

This document provides an overview of the PICU Data Collaborative (PDC) schema specifically for the release provided to teams at the WFPICCS datathon (25-Sept-2026). Please note that this schema is property of the PICU Data Collaborative and provided expressly for the purpose of understanding the released PDC-WFPICCS dataset. Any other use of this schema is prohibited.

Dataset Summary

The PDC-WFPICCS dataset was created as a subset of the full PDC dataset. In both datasets:

- All identifiers are removed or replaced by random, unique identifiers.
- All dates are removed and replaced with durations from a standard time (e.g., seconds since hospital admission).

Specifically, in the PDC-WFPICCS dataset:

- Site numbers are remapped to letter codes. Each letter corresponds to a unique site.
- There are no duplicate ICU admissions within a hospitalization, and no duplicate hospitalizations for a given patient. Therefore, because patient ID : hospitalization ID : ICU admission ID all map to a single value, we have collapsed these identifiers into a `UNIQUE\_ID` identifier. This is a unique key across the `admissions` table and is a foreign key for subsequent tables.
- Subsequent tables include only harmonized data elements (e.g., those mapped to standard PDC names), with the exception of `medications` and `conditionsprocs` which include all rows for the included UNIQUE_ID ICU encounters.
- Data were subsampled from sites such that a holdout set exists within the PDC main dataset. This may be useful if teams are interested in validating any algorithms following the Datathon.

Primary Data

This section provides detailed information on the PDC-WFPICCS schema and data elements. For all tables, columns labelled as Not NULL may not have any NULL values. All other columns are allowed to have NULL values.

1. Admissions

Each row of this table represents one ICU admission within a single hospital admission for a given patient.

Column	Type	Description
unique_id	String	An anonymized, unique identifier for this ICU admission (unique to a single hospitalization and patient)
site_code	String	The letter corresponding to the specific site (e.g., medical center)
icu	String	Location of this ICU admission (PICU, CICU, etc.)
icuadmission	Integer	When patient was admitted to the ICU, measured in seconds since hospital admission
icudischarge	Integer	When patient was (physically) discharged from the ICU, measured in seconds since hospital admission
icumortality	Boolean	0 if survived, 1 if died, null if unknown
sex	String	Male Female Other Unknown
ageatadmission	Integer	Patient's age in days at hospital admission
timeatadmission	Integer	Time of day at hospital admission, measured in seconds since midnight
hospitaldischarge	Integer	When patient was discharged from the hospital, measured in seconds since hospital admission
race	String	Patient race, if available
ethnicity	String	Patient ethnicity, if available
hospitalmortality	Boolean	0 if survived, 1 if died, null if unknown
yearofadmission	Integer	Year of hospital admission

2. ConditionsProcs

Each row in this table represents one diagnosis, procedure, or condition for a patient in the patient population. To ensure completeness of patient histories, we ask that *all* data coded in ICD-9/10 (or other standard nomenclature) for this hospital admission be included, regardless of whether it relates to a specific ICU stay. Conditions and procedures should only be included for hospital admissions (hids) provided in the Hospital Admissions table. Full patient history is not included at this time.

Column	Type	Description
unique_id	String	An anonymized, unique identifier for this ICU admission (unique to a single hospitalization and patient)
recorded	Integer	When this diagnosis/condition was first recorded in the EHR, measured as age in days
nomenclature	String	Which nomenclature standard is used to classify/describe this condition or diagnosis (SNOMED CT, ICD-9, ICD-10, etc.)
code	String	Code for this diagnosis, procedure, or condition concept as defined by the nomenclature
codetext	String	Textual description of diagnosis, procedure, or condition as defined by the nomenclature
onset	Integer	Estimated or actual time of onset, measured as age in days, if available
abatement	Integer	Time of resolution/remission, measured as age in days, if available
abatementboolean	Boolean	1 if the diagnosis or condition is resolved, 0 if unresolved, null if unknown
isprimary	Boolean	1 if this is a primary diagnosis, 0 if secondary, null if unknown

3. Labs

The Labs table records all laboratory test results for the duration of the hospital admission. Each row of this table represents the final result for one laboratory test.

Column	Type	Description
unique_id	String	An anonymized, unique identifier for this ICU admission (unique to a single hospitalization and patient)
nomenclature	String	Nomenclature if nameCode field is taken from a standard such as LOINC, null otherwise
code	String	Code for this type of observation as defined by the nomenclature. If standard nomenclature is unavailable, enter internal ID. Should be consistent across submissions
pdv_name	String	Observation name or title as it appears in the source system or nomenclature, as appropriate
testtype	String	Test or procedure grouper (such as CBC or Liver Function Test) that groups several individual labs (WBC, Hemoglobin, etc.), if appropriate
sampletime	Integer	When the sample was taken, measured in seconds since hospital admission
resulttime	Integer	When the lab result was returned (if known), measured in seconds since hospital admission
pdv_value	String	Actual result
pdv_units	String	Unit of measure for this observation, if appropriate
flag_outofrange	String	True when this value is flagged out of range by harmonization engine, False or Null otherwise
flag_inequality	Integer	True when this value is flagged to contain an inequality by the harmonization engine, False or Null otherwise

4. Vital Signs

The VitalSigns table records vital signs and non-laboratory observations for the patient for the duration of hospital admission.

Column	Type	Description
unique_id	String	An anonymized, unique identifier for this ICU admission (unique to a single hospitalization and patient)
code	String	Code for this type of observation as defined by the source system, if applicable. Should be consistent across submissions
cdc_name	String	Observation name or title as it appears in the source system
starttime	Integer	When this observation began, measured in seconds since hospital admission
endtime	Integer	When this observation ended, measured in seconds since hospital admission, if it had a duration. Null if endTime is not appropriate or not known
cdc_value	String	Actual result
cdc_units	String	Unit of measure for this observation, if appropriate
flag_outofrange	String	True when this value is flagged out of range by harmonization engine, False or Null otherwise
flag_inequality	Integer	True when this value is flagged to contain an inequality by the harmonization engine, False or Null otherwise

5. Interventions

The Interventions table contains interventions performed on the patient. Details regarding complex interventions, such as settings for ventilation, are also included in this table.

Column	Type	Description
unique_id	String	An anonymized, unique identifier for this ICU admission (unique to a single hospitalization and patient)
code	String	Code or identifier for this type of intervention as it appears in the source system. Should be consistent across submissions
cdc_name	String	Procedure name or title as it appears in the source system
starttime	Integer	When this intervention began, measured in seconds since hospital admission
endtime	Integer	When this intervention ended, measured in seconds since hospital admission, if appropriate
cdc_value	String	Result, value, or outcome of this intervention, if appropriate
cdc_units	String	Units associated with the value, if appropriate
detail	String	Additional information required to clarify or interpret this row if name column is not sufficiently detailed. Should only be included when necessary
flag_outofrange	String	True when this value is flagged out of range by harmonization engine, False or Null otherwise
flag_inequality	Integer	True when this value is flagged to contain an inequality by the harmonization engine, False or Null otherwise

6. Medications

The medications table contains all medications administered to ICU patients for the duration of the hospital stay. Each row of this table represents one administration of a single medication.

Column	Type	Description
unique_id	String	An anonymized, unique identifier for this ICU admission (unique to a single hospitalization and patient)
nomenclature	String	Nomenclature if code field is taken from a standard such as SNOMED or , null otherwise
code	String	Code or identifier for this type of medication as it appears in the source system or a standard nomenclature, such as SNOMED-CT
source_name	String	Medication administered to the patient
pdname	String	Harmonized medication name, if it exists
starttime	Integer	When medication administration began for this administration event
endtime	Integer	When medication administration ended for this administration event, if appropriate
route	String	Route or physiologic path of administration of a therapeutic agent into or onto the patient
dosequantity	Double	The volume or amount of medication given at this administration event
quantityunits	String	Unit of measure for the specified dosageQuantity
doserate	Double	For continuous medications, the rate at which the medication was introduced to the patient
rateunits	String	Units for the specified dosageRate
iscont	Boolean	1 if medication is being administered by continuous drip, 0 if intermittent, null if unknown